

Turn your song into a path.

Trail / A guided practice companion that builds short, adaptive sessions from the pieces a musician actually wants to learn.

DESIGN PROPOSAL · NOT A BUILT PRODUCT

The unit of progress is a skill demonstrated in your own music.

For guitarists and bassists who can play but do not always know how to practice. The app supplies a plan without taking away the score.

Bring a piece and a goal. Get a ten-minute session that diagnoses, teaches, checks and schedules your next step.

Trail

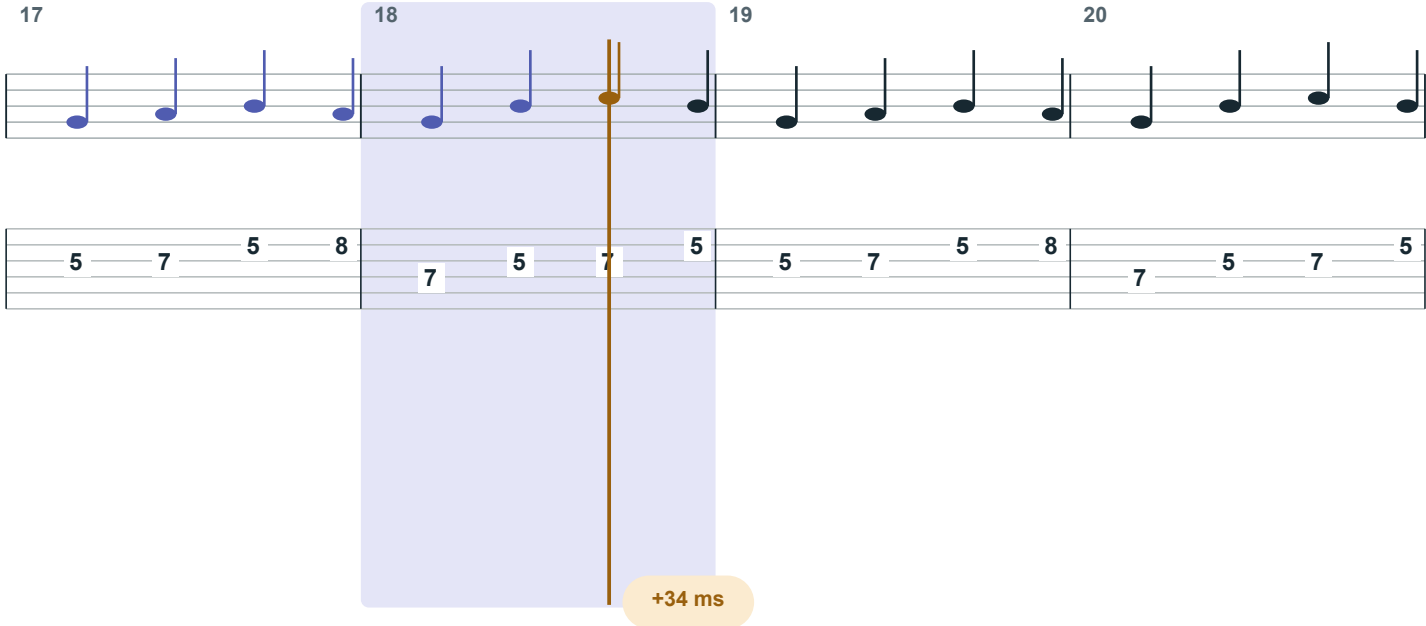
LibraryPracticeJamProgress

Input: ready

Today / make the chorus feel easy

72 BPM

17181920



Play loop

Bars 17–20 • Count-in 1 bar

Direction

A clear next activity reduces the effort of deciding what to practice.

Transfer

Every drill links back to a phrase in the chosen piece.

Memory

A later check distinguishes a temporary success from durable learning.

A complete practice app, with one clear center

Today → Mission → Evidence → Piece; Library, Jam and Skill history remain accessible.

Today plan

Choose 5, 10 or 20 minutes. Balance a recall check, one focus and a return-to-piece performance.

Mission score

Same transport, loop and mixer as a normal score view. The active fragment adds a goal and a concise instruction.

Skill history

Show demonstrated skills with source phrases, tempos and review dates. Separate exercise success from transfer.

Adaptive path

Create a draft phrase and skill map from the imported score. Let users correct it and freely skip any activity.

COMMON CAPABILITIES IN THE FIRST COMPLETE RELEASE

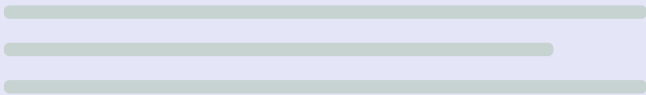
Bring music	Hear and control	Play and improve	Keep and return
GP3–8 + MusicXML import; track and tuning confirmation; score / TAB.	Sampled band; track mute/solo; 40–200% speed; A–B loops; count-in.	Mic / interface input; supported note and timing assessment; evidence-linked drills.	Per-phrase tempo history; reliable-score coverage; saved exercises and jam recipes.
Import report lists degraded or unsupported features. Keep the source file.	Learn mode, assessment mode, mixer, metronome and reference listening.	Clear recording-confidence states; replay; report an incorrect observation.	Guest practice with local saves; optional account sync and deletion controls.

A practice session with a beginning and an outcome

Six frames follow one guitarist or bassist from choosing music to returning with a reason. All interface data are illustrative.

01 Name a goal

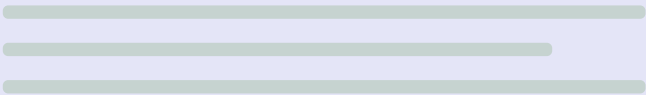
Play this chorus in two weeks



Sam imports a favorite piece and chooses bass, ten minutes per session, and a target tempo. Dates guide the plan; they do not guarantee mastery.

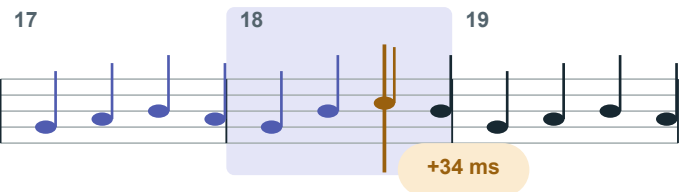
02 Find a starting point

A short, supportive baseline



The app samples two phrases at a comfortable tempo and checks the input. Sam can override the suggested difficulty.

03 Do one focused mission



A compact score shows the exact fragment and the current task. Full-score practice remains accessible.

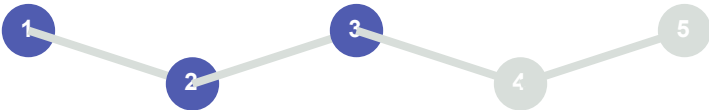
04 See why the plan changes

A specific observation

Notes right; transition late.

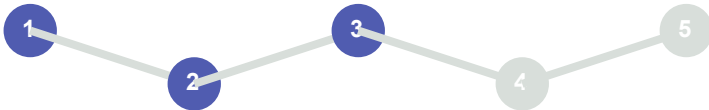
Evidence selects a transition mission. Low-confidence audio instead selects an input check, never a remedial lesson.

05 Use it somewhere new



Sam plays the same rhythm over ii–V–I in A. The app scores the stated exercise goal rather than all improvised notes.

06 Return and remember



Tomorrow begins with a short check of the phrase. A capability badge records where and at what tempo the skill was shown.

The score remains the place where playing happens

Desktop-first concept. Vector mockup uses an invented phrase and illustrative data; notation is schematic.

Trail

LibraryPracticeJamProgress

Input: ready

Today's session

01 Recall

2 min · warm start

02 Transition

5 min · active focus

03 In context

3 min · check

Free practice

Open the whole score

Today / make the chorus feel easy

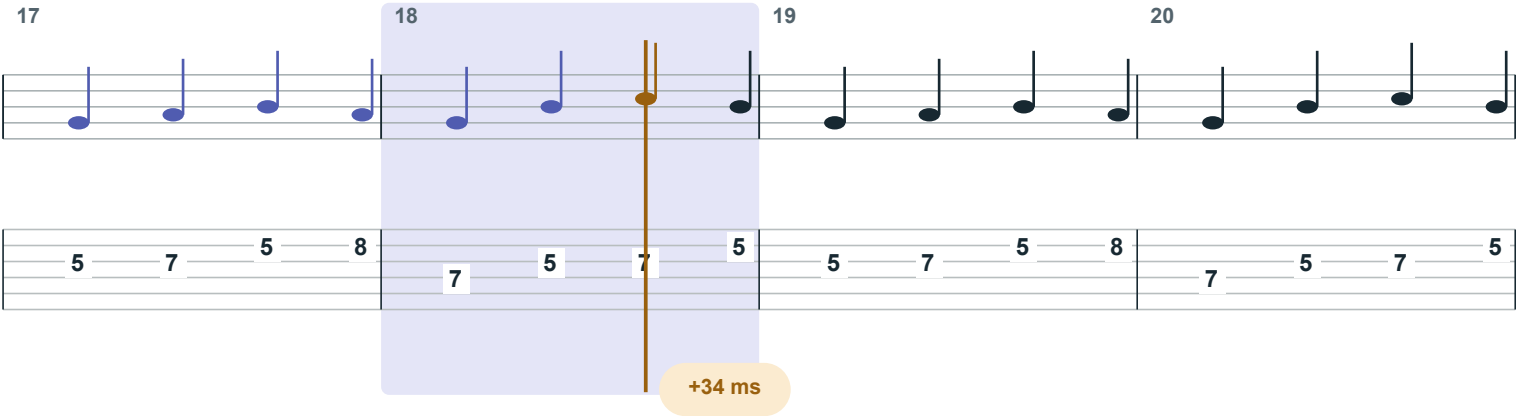
Mission 2 of 3 · Your piece: Evening Study · Original practice sketch

Score + TAB

72 BPM / 90 target

Loop 17–20

17181920



ListenPlay alongCheck take

Pause

- Tempo +

A [17] B [20]

Count-in: 1 barRest: 2 beatsRepeat: 3

Mission: land together

Play the new position on the beat.

First listen, then play. Your next activity depends on this take.

THIS TAKE

Listen for the change

Detailed feedback appears when the loop ends.

BAND MIX

Your part muted

Bass70%Drums55%

Keys35%Click20%

1 / A small plan

The left rail explains session order and remaining time, with an immediate exit to free practice.

2 / A real score

Every mission uses the notation / TAB interface, tempo and loop controls. There is no detached minigame.

3 / An explicit goal

Give one observable task. Update the plan only after a take ends and the evidence is reliable.

PRODUCT EXPLORATION · 29 SEP 2026 · GUITAR + BASS FIRST

04 / 17

Observe → explain → practice → return

Useful feedback needs a visible cause, a manageable next action and a check that the action helped.

Trail

LibraryPracticeJamProgress

Input: ready

You found the notes. Now connect them.

Bars 17–20 · 72 BPM · Take 3

28 / 31
assessed notes matched

26 ms
median [timing]

31 / 32
notes assessable

● Matched ▲ Late × Pitch mismatch ? Unclear

ONE NEXT STEP

The same shift still delays the next beat. Let's isolate it once.

A 90-second mission: hear the transition, play it on one string, then restore the fingering. Finish by playing bars 17–20.

Start the short mission

Replay evidence · Mark incorrect

How the coach chooses this exercise

Use a teacher-authored template catalog: Hear/Repeat, Rhythm Only, Two-Note Shift, Slow-to-Target and Reconnect. Bind templates to score events and a diagnosed issue; record the reason for each choice.

When the recording is unclear

Show “Could not assess this note.” Offer a level or input check; preserve the take without changing mastery.

Reward evidence of learning

Capability badges show what was demonstrated, in which music, at what tempo. A session closes with one improvement and the next review. Cosmetic rewards are optional; practice history never resets.

Trail

LibraryPracticeJamProgress

Input: ready

Skills you can bring back tomorrow.

Same phrase, same rubric. Your last 4 sessions.

YOUR SKILL PATH

1

Recall

Phrase remembered at 64 BPM

2

Connect

Transition demonstrated at 72

3

Transfer

Try the rhythm in a new jam

Tomorrow: one recall check

Timing / steady transitions

Demonstrated at 72 BPM in Evening Study.

Recall check due tomorrow

One short return to the phrase; no daily streak penalty.

Transfer mission unlocked

Try this rhythm over a new backing pattern.

The reward rule

A mission completes only when its stated criterion is met or deliberately self-marked. Verified and self-reported evidence stay distinct. A badge requires an in-piece check; transferable skills require a second context.

Player control

Show “Why this mission?” and “Too easy / too hard / skip.” Never gate access to a piece behind prior missions. Offer guided and free practice from the same screen.

A reason to return

Start with a simple review heuristic: next day, then 3 and 7 days after successful recall. Reschedule from observed performance, and evaluate the heuristic rather than claiming scientific optimality.

Keep comparisons honest

Store tempo, phrase, tuning, rubric, input quality and arrangement version. Show unknown separately; compare like with like.

A plain-language request becomes an editable band

“Shuffle beat ii–V–I in A” is ambiguous. Show the chosen harmony and feel before playback; users can edit either.

TrailLibraryPracticeJamProgressInput: ready

Build a band in a sentence

shuffle beat ii–V–I in A

Interpreted as A major · jazz sevenths · edit any chip

A major

4/4

90 BPM

Shuffle 2:1

8 bars

Jazz trio

Bm7

bars 1–2

E7

bars 3–4

Amaj7

bars 5–8

Drums: shuffle

Bass: walking

Keys: sparse comping

Mission: play roots on beat 1 · Role: bass · Generated bass muted

Preview band

Start practice

Same score, loop and feedback controls

Why it belongs here

Jam is a transfer mission: roots on beat 1, guide tones at changes or a written rhythmic pattern. Label each goal so a creative choice is never automatically counted as an error.

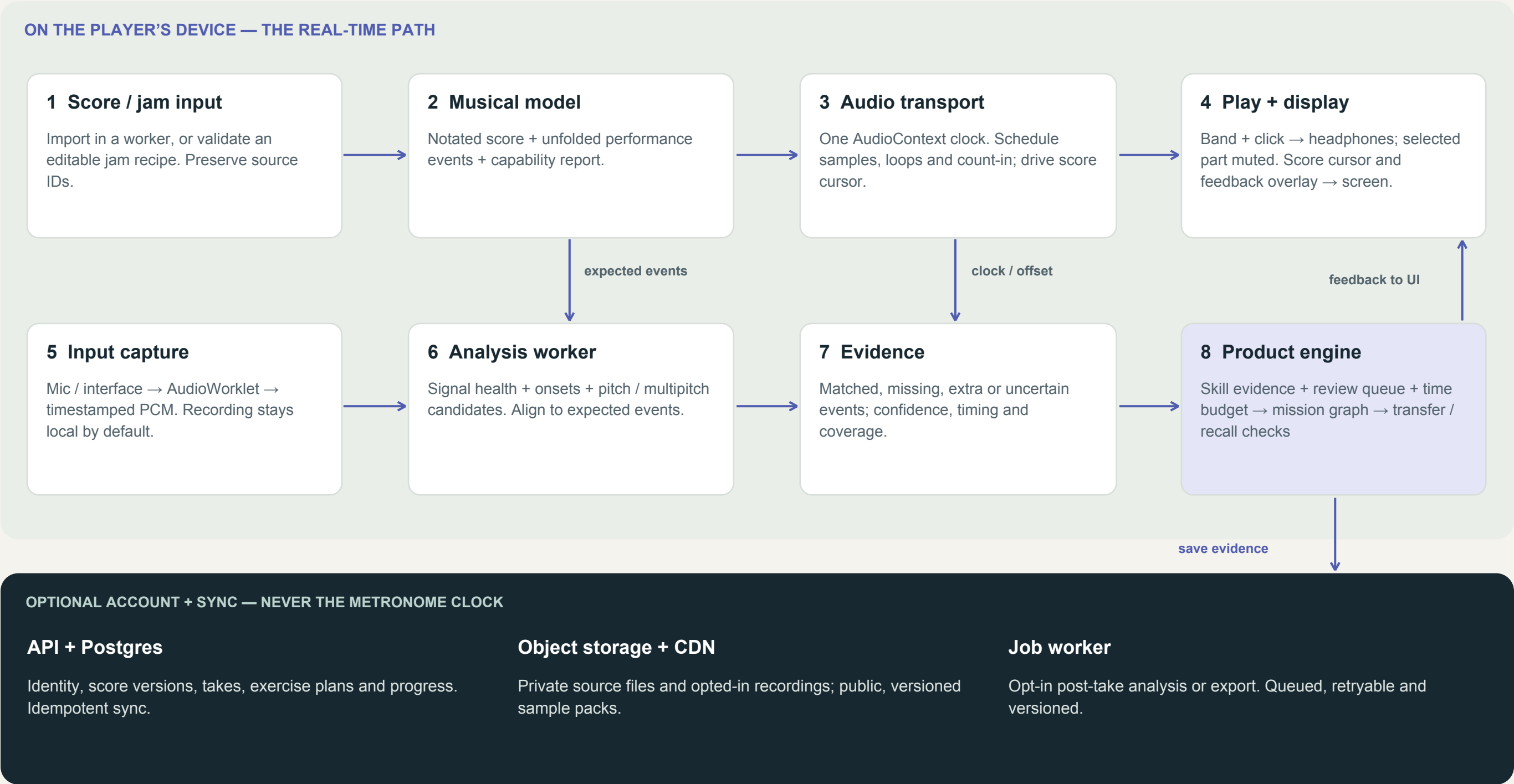
Musical defaults

A major; Bm7 → E7 → Amaj7; 2 + 2 + 4 bars; 4/4 at 90 BPM; eighth-note shuffle 2:1. Offer triads, minor key or dominant blues explicitly.



One musical timeline connects the whole system

Proposed stack: TypeScript + React/Vite; alphaTab adapter; Web Audio; worker-based analysis; a small API and database. [1–8]



The supplied files already reveal the hard cases

Local metadata inspection with alphaTab 1.8.4, plus GP8 XML inspection. Parsing succeeded for all three; rendering and sound were not validated.

Supplied file	Observed structure	What it changes in the design
Born under a bad sign .gp4	6 tracks · 84 notated bars · 90 BPM Bass, guitars, drums, piano and brass.	Start with bass and a selected guitar part. Track names and instrument metadata need user confirmation.
Hotel California .gp4	10 tracks · 121 bars · 74 BPM 122 bent notes; 108 tie destinations.	Large mixer and score layout; sustain and bends. Import emitted custom-bend fallback warnings; inspect those events.
Solo 3 – Comfort Food .gp	GP8.1.2 · 1 track · 53 bars · 88 BPM Embedded MP3; 32 sync points; swing.	Separate notation time from recorded performance time. Blank title metadata needs a filename fallback; preserve tuplets and grace notes.

Import → validate → normalize

1. Check magic bytes, archive size, file count and decompression limits.
2. Parse off the UI thread; disable external XML references.
3. Confirm the playing track, instrument, tuning and capo.
4. Report unsupported notation, sound and grading separately.
5. Persist original bytes + parser version + immutable score version.

A notation model and a performance model

Keep bars, voices, rational durations, rests, ties, ornaments, strings/frets, tuning, capo, written/sounding pitch and source IDs. Unfold repeats into occurrence IDs; map source notes to each occurrence.

Compile expressive timing and tempo into expected events. A tied continuation is not a new attack. Keep MIDI as an export of the richer score model.

Treat the backing band as an instrument

Sample-based playback is part of the core product. Broad format support and beautiful tone are separate acceptance gates. [3–5, 9–10]

Layer	First complete release	Quality and expansion plan
Guitar + bass	Curate or commission clean picked / fingered multisamples with 3+ velocity layers and alternate attacks.	Map tuning, key ranges, release and mute behavior. Add per-voice bends, slides and amp/cab effects; do not assume access to Guitar Pro RSE assets.
Drums + comping	Audition a web-sized Virtuosity Drums subset and VCSL keys; retain source licenses.	Author dynamics, choke groups, bass voice leading and sparse comping. Commission missing core sounds with web redistribution rights.
Pack pipeline	Offline tooling trims, normalizes and maps licensed WAVs into manifests, zones and compressed assets.	Load active tracks first; decode before playback; cache versioned packs. Target a 15–40 MB starter download and ≤256 MB decoded samples on the baseline laptop; validate with real packs.
Fallback sounds	A permitted GM SoundFont covers unusual instruments; label it as standard sound quality.	Quality packs must pass listening tests before launch. The fallback is continuity during downloads, not the premium-sound acceptance criterion.

Transport owns musical time

One AudioContext and event scheduler drive playback. Prototype alphaTab’s IExternalMediaHandler for cursor sync; retain a bounds-based cursor fallback. Tempo scales event time, not sample pitch. Cancel events on seek; handle sustain, count-in, loop tails and tied notes explicitly.

Imported recordings have their own path

Offer Reference recording and Generated band modes. GP8 sync maps align audio to bars. Fixed recordings need pitch-preserving time-stretching and cannot mute a single part without stems. Evaluate a licensed DSP implementation; avoid two independent playback clocks.

Generate musical events, then play them beautifully

A constrained arranger is an achievable first product. A language model may help interpret wording later; it never owns the timing clock or the grading target.

VALIDATED JAM RECIPE

Key / mode	A / major
Harmony	Bm7 × 2 E7 × 2 Amaj7 × 4
Meter / tempo	4/4 / 90 BPM
Groove	shuffle-eighths / ratio 2:1
Band	drums + bass + piano
Player role	guitar
Form / variation	8 bars / seed 1042
Assessment	free, constrained or written

From words to an arrangement you can inspect

1. A grammar recognizes key, Roman numerals/chord names, feel, meter, tempo and bar counts. Unknown terms stay highlighted; the structured form always works.
2. Normalize chord spelling and mode; validate durations and instrument ranges. Show defaults rather than silently guessing.
3. A style template chooses drum cells, bass patterns and comping density. Constrained voice leading keeps chord shapes playable and bass movement idiomatic.
4. Emit note, articulation and dynamic events plus an editable score/chord view. Save the recipe, generator version and seed.

Make the groove intentional

For straight eighths, an offbeat is at 1/2 beat; a 2:1 shuffle places it at 2/3. Swing only eligible pairs, not explicit tuplets or imported timing twice.

Drums use authored accents, alternate samples and choke groups. Bass can use roots, fifths and reviewed approach-note rules. Comping selects sparse chord voicings within an instrument range. Humanization is seeded and bounded; store the exact reference events.

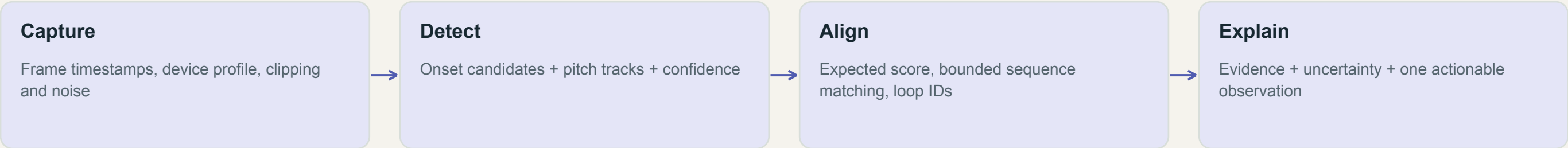
Connect the band to practice

Mute the role the user will play, while retaining its expected events for a written or constrained task. In open improvisation, keep harmony visible and offer observations rather than a pitch grade.

Preview a 4–8-bar form, loop it, change key or tempo, and save it. Regenerate only at a stopped state or next form boundary. A generated arrangement enters the same transport, score, assessment and history pipeline as an imported piece.

Know what was played before judging it

Proposed capability tiers, not claims of proven accuracy. Ship each assessed mode only after it passes an instrument- and input-specific benchmark. [5–8]



Playing mode	Assessment method	What the product promises
Clean single notes Initial grading mode	Spectral-flux onsets and YIN-style pitch candidates; bounded sequence alignment with match, missing and extra events. Retain detected pitches before consulting the score.	Notes and timing on validated clean guitar / bass ranges. Target B0–E6, with longer windows for low bass; validate the range before advertising it.
Chords and strumming Post-take extension	Multipitch model candidate (Basic Pitch) + score-conditioned event matching; group staggered string attacks.	Grade rhythm first; enable note-set checks only on validated chord families. Distortion, bleed and ambiguous voicings can remain ungraded.
Bends, slides, legato Technique extension	Compare pitch contours and target arrival. Model ties and hammer-ons separately from picked onsets.	Show expressive notes in notation and playback immediately. Add technique-specific grading after musician-labeled validation; audio cannot prove fingering.
Free improvisation Jam mode	Beat-relative onset and phrase observations; optional announced constraints such as roots on beat 1.	No pitch-correctness score without a defined target. Musical taste, tone and creative choices are not reduced to a universal percentage.

LIVE AND FINAL FEEDBACK HAVE DIFFERENT JOBS

Live markers are provisional. Reconcile each take in a worker; preserve evidence and rubric versions. Keep prior results if models change.

A score is only useful when its meaning is clear

Store raw counts, timing distributions, coverage and rubric version. The following thresholds are starting hypotheses for teacher validation.

Separate pitch, timing and uncertainty

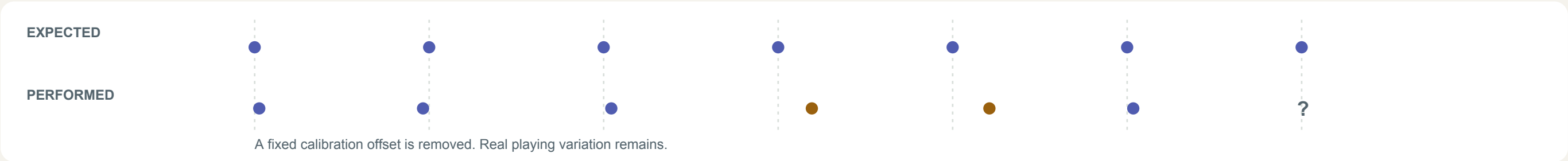
Pitch: correct expected notes / assessable expected notes; also show missing notes and extra attacks. Coverage: assessable expected notes / all eligible expected notes.

Timing: signed median, median absolute onset error and the 90th percentile on matched onsets. Preserve the bias: consistent lateness must not be automatically centered away.

Keep the task stable during assessment

Bind every take to a track, phrase occurrence, tempo map, tuning, input profile, score version and rubric. Backing remains fixed while assessing; bounded alignment may recover the score location without forgiving actual timing errors.

Distinguish “missed” from “not assessable.” Silence on a healthy signal can be a miss; clipping or ambiguous pitch can be unknown.



Calibrate the recording and playback path

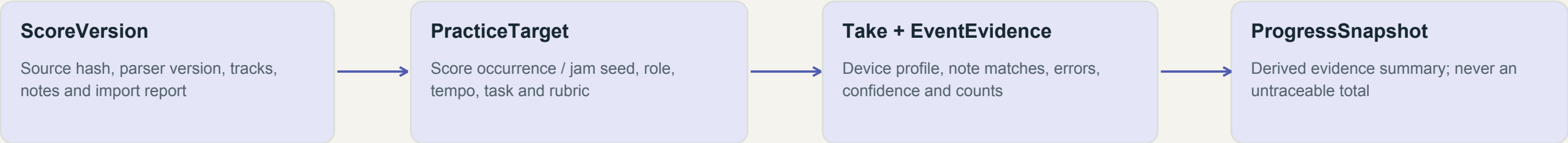
Measure round-trip delay with loopback where possible. A guided musical alignment is provisional because it includes human timing. Save uncertainty per device / sample rate; never recalibrate from each scored take. Recheck after route changes; unreliable Bluetooth timing disables fine timing grades.

A proposed tempo ladder

For a simple picked-note drill: ≥90% note matches, ≥90% coverage, extras ≤5%, median |onset| ≤30 ms and p90 ≤60 ms. Require two qualified takes; offer +4 BPM next loop. Two difficult takes suggest −8 BPM or a smaller phrase. Retain only after a later check; adapt tolerances for expressive tasks.

Save the evidence that makes progress reproducible

Start with a modular application and one backend. Keep sound scheduling and the initial feedback loop independent of network access.



Session planner

SkillEvidence {skillTag, sourceTake, context, tempo, confidence}; Mission {templateVersion, eventRange, target, reason}; ReviewDue {itemId, dueAt}.

POST /plans/today; POST /missions/:id/complete; GET /reviews/due. Replaying a saved plan keeps the exact template and rubric versions.

Local first, account optional

IndexedDB stores scores, takes and a sync outbox. UUIDs and idempotency keys prevent duplicate attempts. Immutable score and rubric versions avoid corrupting old progress. Optional account sync uses an authenticated API, Postgres and private object storage with short-lived URLs.

API and job boundaries

POST /scores creates a version after upload validation; POST /takes records a completed attempt; GET /progress filters comparable evidence. A background job can refine an opted-in recording; results carry model and rubric versions.

Use a queue only for post-take work. Signed uploads, ownership checks, quotas and archive limits protect the shared backend.

Privacy, recovery and release operations

No raw recording upload by default. Explicitly save or upload a take; offer per-take and account deletion plus export. Recordings are not model-training data without separate opt-in.

Show local / syncing / saved / failed states. Retry safely after offline use. Monitor input failure, analysis time, underruns and disputed feedback; exclude private score/audio content from telemetry.

A gated path to a complete Trail beta

Planning estimate: 20–26 weeks with three engineers, a half-time designer, fractional QA and a musician/teacher partner. Ranges are assumptions, not vendor estimates.

Phase / duration	Primary owner	Deliverable and exit criterion
0 · Prove the risks 2 weeks	Audio/DSP + frontend	Render and audition the three supplied files; label input recordings; verify clock and external-cursor integration. Exit: fidelity report, sample rights inventory and a measurable assessment baseline.
1 · Complete playback 3 weeks	Frontend + audio	Import report, score/TAB, selected track, loop, speed, count-in and mixer. Canonical events handle repeats and tempo changes. Exit: deterministic playback and cursor tests; seek and loop sustain behavior checked.
2 · Hear the student 4–6 weeks	DSP + platform + sound contractor	Capture, device checks, alignment, monophonic evaluation and evidence review. In parallel, curate quality sounds. Exit: precision, coverage, latency and blind-listening gates pass on supported setups.
3 · Deliver the concept 6–8 weeks	Product + full team	Build template bindings, editable phrase/skill map, time-boxed session planner, mission state and review queue. Exit: a mission improves the linked phrase and a user can explain the recommendation.
4 · Finish the whole loop 3–4 weeks	Full stack + product	Add or complete jam recipes, arrangement-to-score, account sync, progress, retention checks and recovery states. GP8 reference audio uses a validated sync/time-stretch path. Exit: full user journey works across reloads.
5 · Pilot and harden 2–3 weeks	Full team + teacher + QA	Run the musician pilot; test Safari/Chromium, long pieces and input changes. Audit accessibility, storage, sample licenses and feedback disputes. Exit: launch gates met and supported modes published.

After the complete beta

Budget 8–14 more weeks for validated chord families, bend/slide assessment and more styles; research may extend this. New instruments require new profiles and benchmarks. Core note, timing, training, progress and jam workflows already exist in the beta.

Cost drivers

Three engineers dominate cost. Allow \$10k–\$30k for sound/style work; obtain quotes after auditioning the first pack.

Define the promises, then test them

All numeric gates below are proposed acceptance targets. None was measured on a finished application during this design exercise.

Risk / requirement	Evidence needed before release
Import and notation fidelity	Compare GP3/4/5/6/7/8 and MusicXML fixtures to trusted notation and MIDI references. Include tuning, capo, repeats, alternate endings, tempo/meter changes, ties, swing and tuplets. The three local files alone are insufficient.
Trustworthy feedback	Collect consented clean DI and mic takes from guitarists and bassists, plus noisy/distorted negatives. Split by player and instrument. On supported cases: $\leq 2\%$ false mistake flags among teacher-labeled correct events and $\geq 90\%$ coverage and $\geq 85\%$ recall of labeled mistakes; report all three together.
Clock and response	Test injected timing offsets and loopback recordings. Target ≤ 10 ms p95 residual error on calibrated wired setups; onset markers ≤ 150 ms p95, final review ≤ 5 s for a 30 s mono take on a stated midrange laptop. Low bass pitch may appear later.
Sound and reliability	Blind-listen with ≥ 12 musicians; target ≥ 8 preferring the quality band over fallback. Test 30-minute sessions, seeks and tempo changes without stuck notes or underruns on the baseline device. Preload and cache only the needed patches.
Learning and usability	Recruit 12 players who report unstructured practice. Compare self-chosen looping with guided ten-minute sessions on matched excerpts. Recheck after 24–72 hours and ask whether guidance felt useful or restrictive.
Concept-specific failure	Generated lessons can feel generic or patronizing. Test user edits to the inferred skill map and whether the chosen exercise actually transfers back to the piece.

Release scope covers every requested workflow

Upload → score/TAB → sampled playback → tempo/loop → live audio → supported note/timing evidence → generated exercise → progress history. Typed jam request → editable form → sampled backing → the same practice and review tools.

Broader validation

Keyboard controls, text alongside colors, large targets, reduced motion and a tablet score view. Phone gets a focused single-system view.

What is verified, what is proposed, what comes next

External technical sources reviewed September 29, 2026. Links are clickable. Product behavior, thresholds, schedules and budgets are design proposals.

[1] [alphaTab format catalog](#)

Import candidates: GP3–8 and MusicXML. Verify each feature on a fixture corpus.

[2] [alphaTab GP8 compatibility](#)

Reading, engraving and audio support differ; importing a feature is not proof of fidelity.

[3] [alphaTab external audio synchronization](#)

External playback hooks and GP8 sync data. Built-in backing and synthesis are alternative modes.

[4] [alphaTab project / licensing](#)

MPL-2.0; notation, interactive playback and SoundFont support. Review shipped dependencies and notices.

[5] [W3C Web Audio specification](#)

AudioContext, AudioWorklet and output timestamps underpin the proposed timing pipeline.

[6] [W3C Media Capture and Streams](#)

Input permissions and audio constraints. Inspect actual settings; requests are not guarantees.

[7] [Spotify Basic Pitch repository](#)

Candidate for post-take multipitch analysis. Best with one instrument; benchmark before adopting.

[8] [Basic Pitch research paper](#)

Bittner et al., 2022. Research basis, not evidence of student-assessment accuracy.

[9] [Versilian Community Sample Library](#)

CC0 library candidate for selected accompaniment sounds; audition and curate per patch.

[10] [Virtuosity Drums source and license](#)

Candidate drum source. Keep the exact pack license and provenance with derivative assets.

LOCAL EVIDENCE

Source files: main checkout /guitarpro; metadata and import warnings recorded in research/file-analysis.json. Prototype reviewed: timingAnalyzer.ts, microphoneRecorder.ts, metronome.ts, sampler.ts and pitch.ts. Existing onset logic and UI ideas are references; this plan does not depend on keeping their implementation.

DECISION THIS DESIGN SUPPORTS

Best if the central problem is “I do not know what to practice next.” Requires stronger pedagogy and more content operations than Phrase.